

Analysis note: charged-particle multi-particle cumulants in ALEPH hadronic e^+e^- collisions at the Z pole

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Abstract

This note documents a first charged-particle multi-particle cumulant analysis of archived ALEPH hadronic e^+e^- collisions at $\sqrt{s} \simeq 91.2$ GeV. The analysis follows the public Electron-Positron Alliance note style: the main body defines the observable, data setup, event and track selections, cumulant calculation, statistical uncertainty treatment, and the current preliminary results. The reported observables are $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity, evaluated with respect to both the laboratory beam axis and the event thrust axis. The cumulant implementation uses exact ordered, distinct-particle correlations through eighth order and propagates statistical uncertainties with delete-one-chunk jackknife resampling over sixteen event chunks. The present result is a detector-level charged-particle study; detector-response, acceptance, and model-dependent systematic uncertainties are not yet included. A matched 1994 reconstructed-MC comparison is included as a baseline closure-style check of the beam-axis and thrust-axis trends.

Keywords: ALEPH, archived e^+e^- data, charged-particle cumulants, flow harmonics, Q-cumulants

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1 Introduction

Long-range azimuthal correlations and multi-particle cumulants are central tools for separating collective and non-collective correlation structures in high-energy collisions. In hadronic and nuclear collisions, the hierarchy of $v_n\{2\}$, $v_n\{4\}$, $v_n\{6\}$, and $v_n\{8\}$ provides a compact way to test whether the measured azimuthal anisotropy is dominated by a common event-level geometry or by few-particle correlations. The same techniques can be applied to archived e^+e^- data, where the absence of beam remnants and a well-defined color-singlet initial state provide a clean reference for jet fragmentation, parton-shower, and hadronization effects.

The starting point for this analysis is the ALEPH StudyMult charged-particle workflow and the previous ALEPH two-particle-correlation analysis of archived LEP data [1]. The present note extends that infrastructure from two-particle Fourier coefficients to exact multi-particle cumulants through eight particles. The implementation follows the Q-cumulant formalism of Refs. [2, 3] and the cumulant combinations used in small-system flow measurements, including the CMS high-multiplicity pp reference analysis [4].

This document intentionally follows the compact public-note format used in recent Electron-Positron Alliance analysis notes. The main body contains the observable definition, active data setup, analysis method, validation, and preliminary results, while detailed corrections and systematic variations are identified as future work rather than folded into the present detector-level result.

2 Observable Definition

For each selected event, charged particles are represented by their azimuthal angles ϕ_i in a chosen coordinate system. The present study uses two coordinate definitions:

- beam axis: ϕ_i is the laboratory azimuth around the ALEPH beam axis;
- thrust axis: the event is rotated into a frame defined by the event thrust axis, and ϕ_i is the azimuth around that axis.

The harmonic used in this note is $n = 2$. For an event with multiplicity M , the ordered distinct-particle $2k$ -particle correlation is

$$\langle 2k \rangle_n = \frac{1}{M(M-1)\cdots(M-2k+1)} \sum_{\substack{i_1, \dots, i_{2k} \\ \text{all distinct}}} \exp \left[in (\phi_{i_1} + \cdots + \phi_{i_k} - \phi_{i_{k+1}} - \cdots - \phi_{i_{2k}}) \right]. \quad (1)$$

The event-ensemble averages of these quantities are accumulated with their natural ordered-tuple weights. The cumulants are then

$$c_2\{2\} = \langle 2 \rangle, \quad (2)$$

$$c_2\{4\} = \langle 4 \rangle - 2 \langle 2 \rangle^2, \quad (3)$$

$$c_2\{6\} = \langle 6 \rangle - 9 \langle 4 \rangle \langle 2 \rangle + 12 \langle 2 \rangle^3, \quad (4)$$

$$c_2\{8\} = \langle 8 \rangle - 16 \langle 6 \rangle \langle 2 \rangle - 18 \langle 4 \rangle^2 + 144 \langle 4 \rangle \langle 2 \rangle^2 - 144 \langle 2 \rangle^4. \quad (5)$$

The corresponding flow-like coefficients are defined as

$$v_2\{2\} = \sqrt{c_2\{2\}}, \quad (6)$$

$$v_2\{4\} = [-c_2\{4\}]^{1/4}, \quad (7)$$

$$v_2\{6\} = [c_2\{6\}/4]^{1/6}, \quad (8)$$

$$v_2\{8\} = [-c_2\{8\}/33]^{1/8}. \quad (9)$$

If the required cumulant sign is not physical for a given bin, the corresponding $v_2\{m\}$ point is suppressed in the plotted result and is stored as zero in the exported CSV table.

3 Data Setup and Selections

The current production uses the local StudyMult-format ALEPH file

`/data/yjlee/StudyMult/DataFiles/ROOTfiles/cleaned_ALEPH_Data-all.aleph.root.`

This file contains 59,874 selected hadronic events in the tree `t`. The input format is the StudyMult fixed-array layout with `nParticle`, `px`, `py`, `pz`, and `pwflag` branches.

Charged particles are selected using the StudyMult charged-particle convention

$$\text{pwflag} \in \{0, 1, 2\}, \quad (10)$$

with $p_T > 0.4$ GeV for both the beam-axis and thrust-axis cumulants. The event thrust axis is computed from reconstructed particle momenta, following the local ALEPH StudyMult convention used by the prototype analyses. The charged-particle multiplicity bins used in this note are

$$[0, 10), [10, 15), [15, 20), [20, 25), [25, 30), [30, 35), [35, 40), [40, \infty).$$

The corresponding event counts are shown in Table 1.

Table 1: Selected event counts in the multiplicity bins used for the current detector-level cumulant analysis.

Multiplicity bin	Events
[0, 10)	7483
[10, 15)	17665
[15, 20)	16856
[20, 25)	9452
[25, 30)	4753
[30, 35)	2287
[35, 40)	955
[40, ∞)	423

4 Cumulant Implementation

The analysis code computes the ordered correlations in Eq. (1) using an exact elementary-symmetric polynomial recursion over the event-level phase factors $\exp(in\phi_i)$. This avoids self-correlations by construction and uses the same ordered-particle denominator as the Q-cumulant definition. The recursion is applied for 2, 4, 6, and 8 particles in each multiplicity bin and for both axis definitions.

The implementation was checked in two ways. First, the code contains a `--SelfTest 1` mode that compares the production correlator against brute-force ordered distinct-particle sums for low-multiplicity toy events. Second, the cumulant combinations and $v_2\{m\}$ normalizations were compared against the published Q-cumulant formulae in Refs. [2, 3, 4]. The signs and normalization factors in the cumulant and $v_2\{m\}$ definitions above match those references.

5 Statistical Uncertainties

The production job was split into sixteen contiguous event chunks. Each chunk stores independent numerator and denominator sums for every correlator and multiplicity bin. The summary step first merges all chunks to obtain the central value and then recomputes the full nonlinear cumulant chain after removing one chunk at a time. The quoted statistical uncertainty is the delete-one-chunk jackknife estimate

$$\sigma_{\text{JK}} = \sqrt{\frac{N_{\text{chunk}} - 1}{N_{\text{chunk}}} \sum_{j=1}^{N_{\text{chunk}}} (x_{(j)} - \bar{x})^2}, \quad (11)$$

where $x_{(j)}$ is the result with chunk j removed. This propagates the nonlinear transformations from raw correlations to cumulants and then to $v_2\{m\}$. The procedure assumes the chunks are statistically independent; if the input file ordering tracks detector running conditions, the jackknife may also contain a small run-block component.

6 Preliminary Results

Figures 1 and 2 show the current detector-level $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ results versus charged-particle multiplicity. In both coordinate systems the two-particle value is larger than the higher-order values, as expected when few-particle and jet-like correlations contribute. The higher-order beam-axis points remain numerically stable through most bins, while some high-order thrust-axis and high-multiplicity points become sign-unstable or carry large jackknife uncertainties because the event counts and ordered eight-particle statistics are limited in those bins.

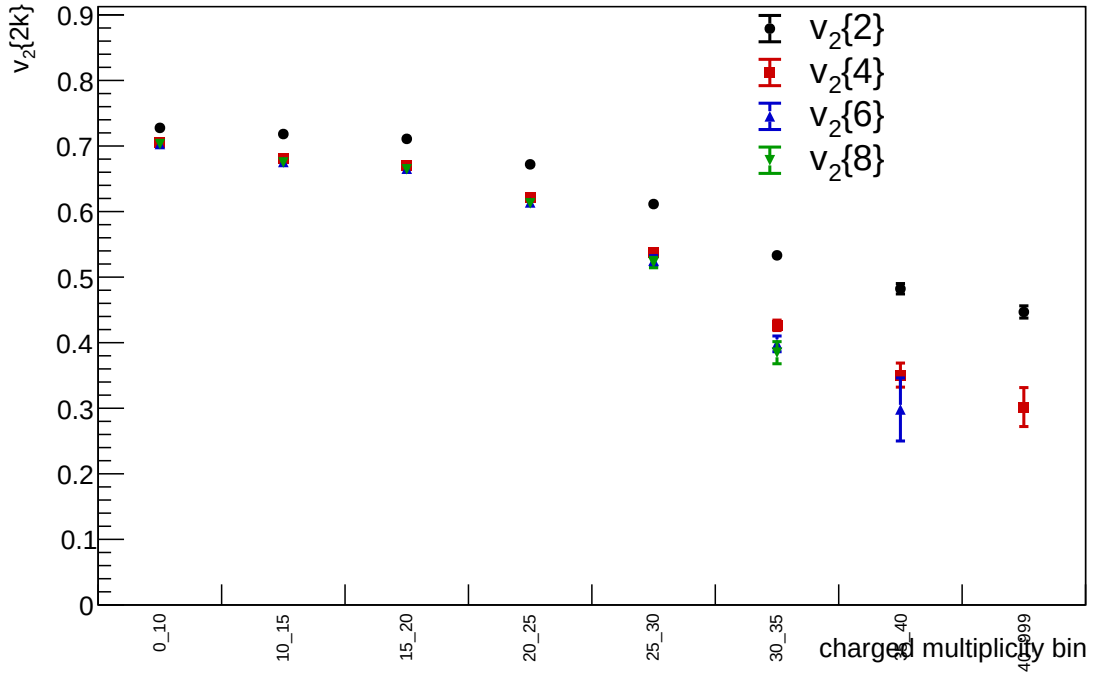


Figure 1: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using the laboratory beam-axis azimuth. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks. Missing high-order points indicate cumulant signs that do not give a real-valued $v_2\{m\}$ in the current sample.

6.1 Matched 1994 reconstructed-MC comparison

A separate MC production was run on the matched LEP1 1994 reconstructed-MC sample

```
/data/yjlee/ALEPH_Agentic_Event_Shape_Analysis/DataProcessing/temp/alephMCRecoAfterCutPa
ths_1994_thrust_pt04_t.root.
```

For a like-for-like comparison, the corresponding 1994 data sample

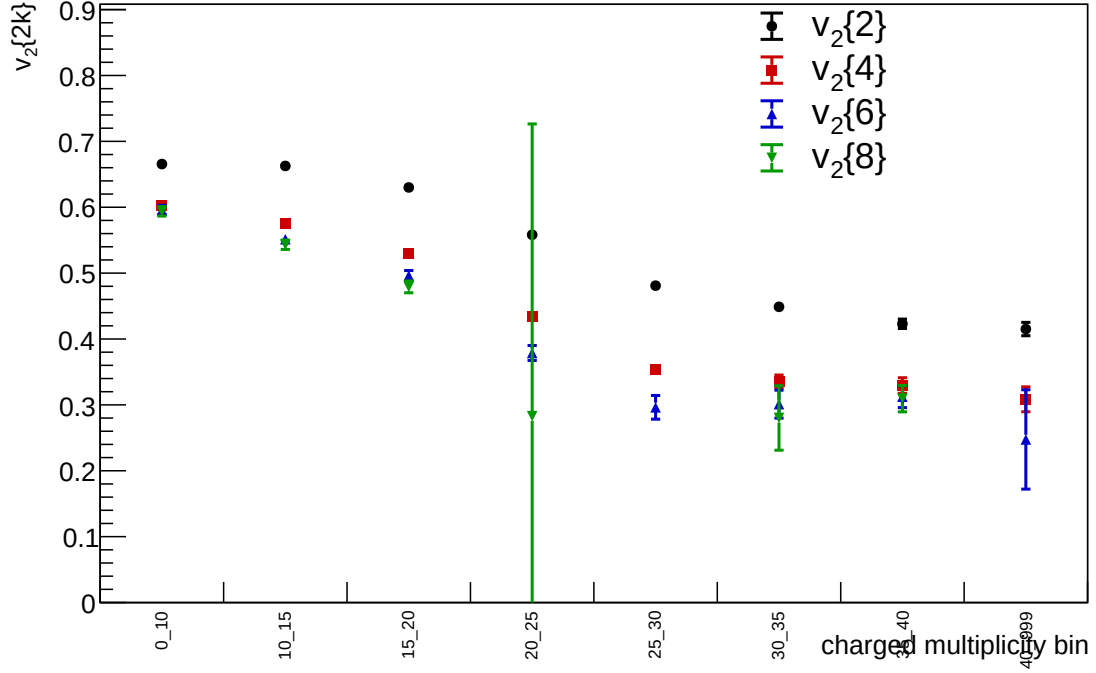


Figure 2: Detector-level charged-particle $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ as functions of selected charged-particle multiplicity using azimuth around the event thrust axis. Error bars are delete-one-chunk jackknife statistical uncertainties from sixteen event chunks.

Table 2: Numerical beam-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as “–” are sign-invalid or unavailable in the current sample.

Multiplicity	$v_2\{2\}$	$v_2\{4\}$	$v_2\{6\}$	$v_2\{8\}$
[0, 10)	0.7277 ± 0.0024	0.7047 ± 0.0032	0.7031 ± 0.0034	0.7029 ± 0.0034
[10, 15)	0.7182 ± 0.0027	0.6802 ± 0.0040	0.6757 ± 0.0041	0.6748 ± 0.0042
[15, 20)	0.7109 ± 0.0023	0.6705 ± 0.0038	0.6654 ± 0.0041	0.6645 ± 0.0042
[20, 25)	0.6720 ± 0.0029	0.6210 ± 0.0045	0.6140 ± 0.0049	0.6125 ± 0.0050
[25, 30)	0.6114 ± 0.0044	0.5372 ± 0.0070	0.5251 ± 0.0077	0.5222 ± 0.0079
[30, 35)	0.5332 ± 0.0046	0.4265 ± 0.0080	0.3982 ± 0.0120	0.3847 ± 0.0169
[35, 40)	0.4823 ± 0.0079	0.3506 ± 0.0184	0.2981 ± 0.0481	–
[40, ∞)	0.4469 ± 0.0093	0.3018 ± 0.0297	–	–

Table 3: Numerical thrust-axis $v_2\{m\}$ values. Uncertainties are jackknife statistical uncertainties. Entries shown as “–” are sign-invalid or unavailable in the current sample.

Multiplicity	$v_2\{2\}$	$v_2\{4\}$	$v_2\{6\}$	$v_2\{8\}$
[0, 10)	0.6655 ± 0.0045	0.6034 ± 0.0068	0.5958 ± 0.0070	0.5938 ± 0.0072
[10, 15)	0.6627 ± 0.0039	0.5748 ± 0.0056	0.5515 ± 0.0065	0.5431 ± 0.0071
[15, 20)	0.6300 ± 0.0051	0.5294 ± 0.0068	0.4962 ± 0.0078	0.4791 ± 0.0090
[20, 25)	0.5582 ± 0.0046	0.4341 ± 0.0063	0.3787 ± 0.0114	0.2828 ± 0.4437
[25, 30)	0.4810 ± 0.0025	0.3535 ± 0.0066	0.2963 ± 0.0180	–
[30, 35)	0.4488 ± 0.0050	0.3359 ± 0.0094	0.3011 ± 0.0212	0.2802 ± 0.0490
[35, 40)	0.4231 ± 0.0071	0.3294 ± 0.0118	0.3125 ± 0.0166	0.3094 ± 0.0200
[40, ∞)	0.4151 ± 0.0101	0.3084 ± 0.0189	0.2475 ± 0.0754	–

/data/yjlee/ALEPH_Agentic_Event_Shape_Analysis/DataProcessing/temp/LEP1Data1994_recons_aftercut-MERGED_thrust_pt04_t.root

was processed with the same cumulant settings, multiplicity bins, and sixteen-chunk jackknife procedure. The processed input contains 771,597 MC entries and 1,365,440 data entries. These samples are separate from the all-StudyMult data-only production shown above, and are used only for the reconstructed-level data/MC comparison in this subsection.

Figures 3 and 4 show the standalone MC result for the beam and thrust axes. The beam-axis MC follows the same broad multiplicity dependence as the data-only result: the cumulant hierarchy is tight at low multiplicity and decreases with increasing multiplicity. In the thrust axis, the two-particle coefficient rises from low to intermediate multiplicity, while several low-multiplicity higher-order coefficients are sign-invalid and therefore not plotted.

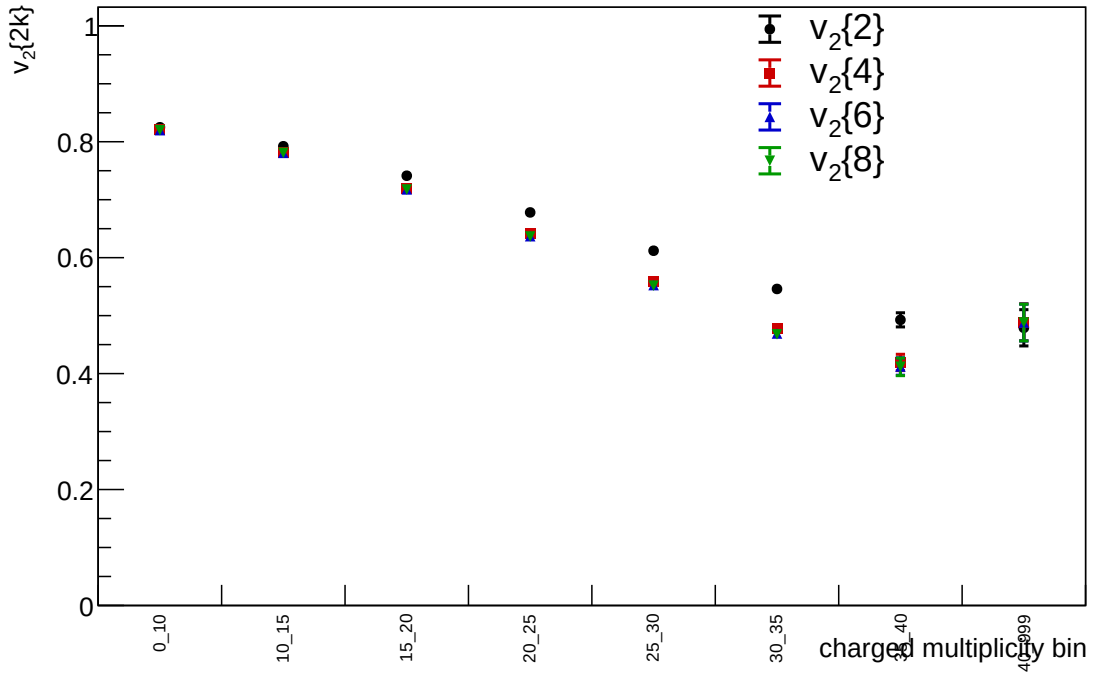


Figure 3: Matched LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using the laboratory beam-axis azimuth.

Figures 5 and 6 compare the matched 1994 data and reconstructed MC directly. For the beam axis, the data/MC agreement is at the percent level over most of the populated multiplicity range; the largest deviations appear in the highest-multiplicity bin, where the statistical uncertainty is also largest. For the thrust axis, $v_2\{2\}$ is well reproduced through intermediate multiplicity. The higher-order thrust-axis coefficients are more sensitive to cumulant sign changes and limited statistics, so missing points indicate bins where either data or MC did not yield a real value under the standard cumulant sign convention.

7 Current Limitations

The present note should be read as a detector-level analysis milestone, not as a final corrected physics result. The main limitations are:

- no detector-response or acceptance correction is applied to the cumulants;

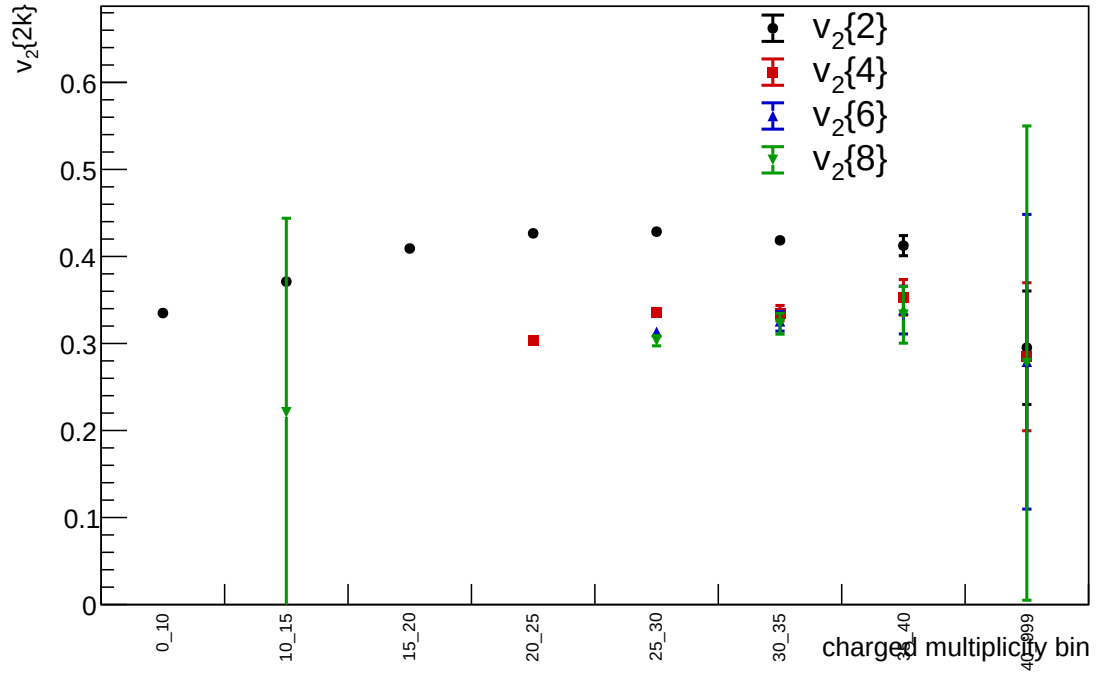


Figure 4: Matched LEP1 1994 reconstructed-MC $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus selected charged-particle multiplicity using azimuth around the event thrust axis.

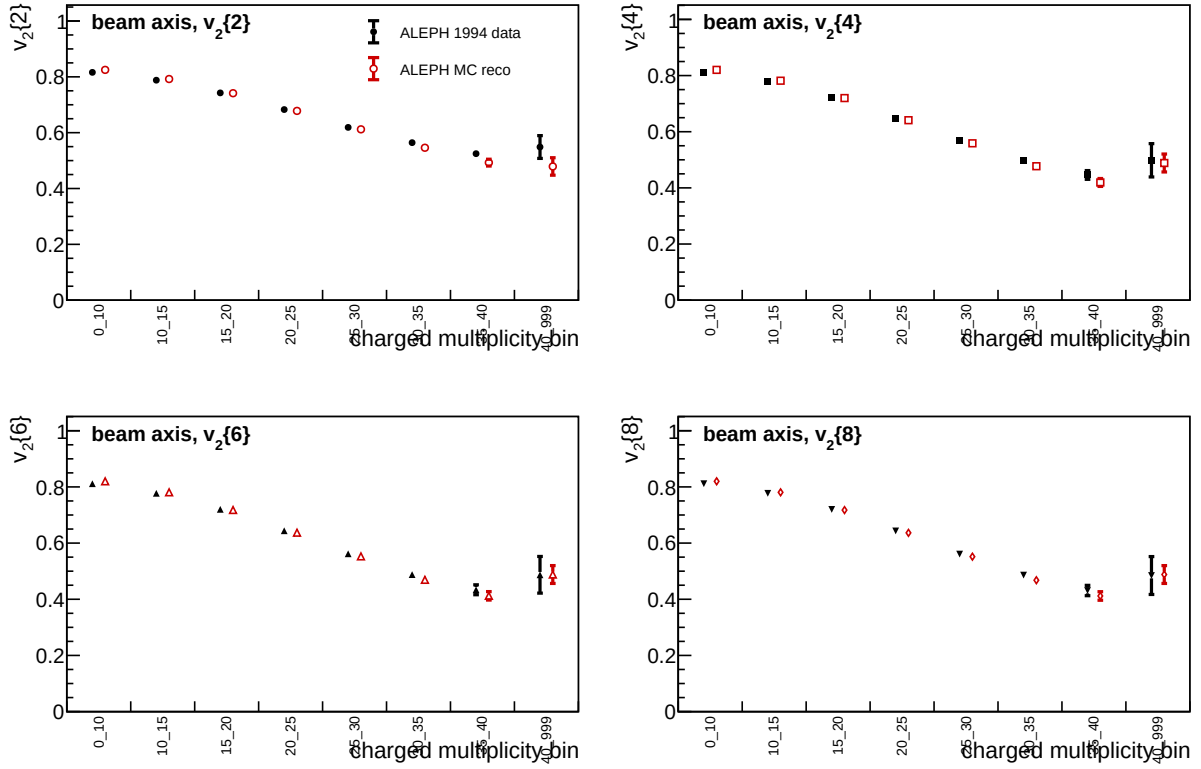


Figure 5: Beam-axis comparison of matched LEP1 1994 data and reconstructed MC. Each panel shows one cumulant order, with data and MC offset slightly in the charged-multiplicity bin for readability.

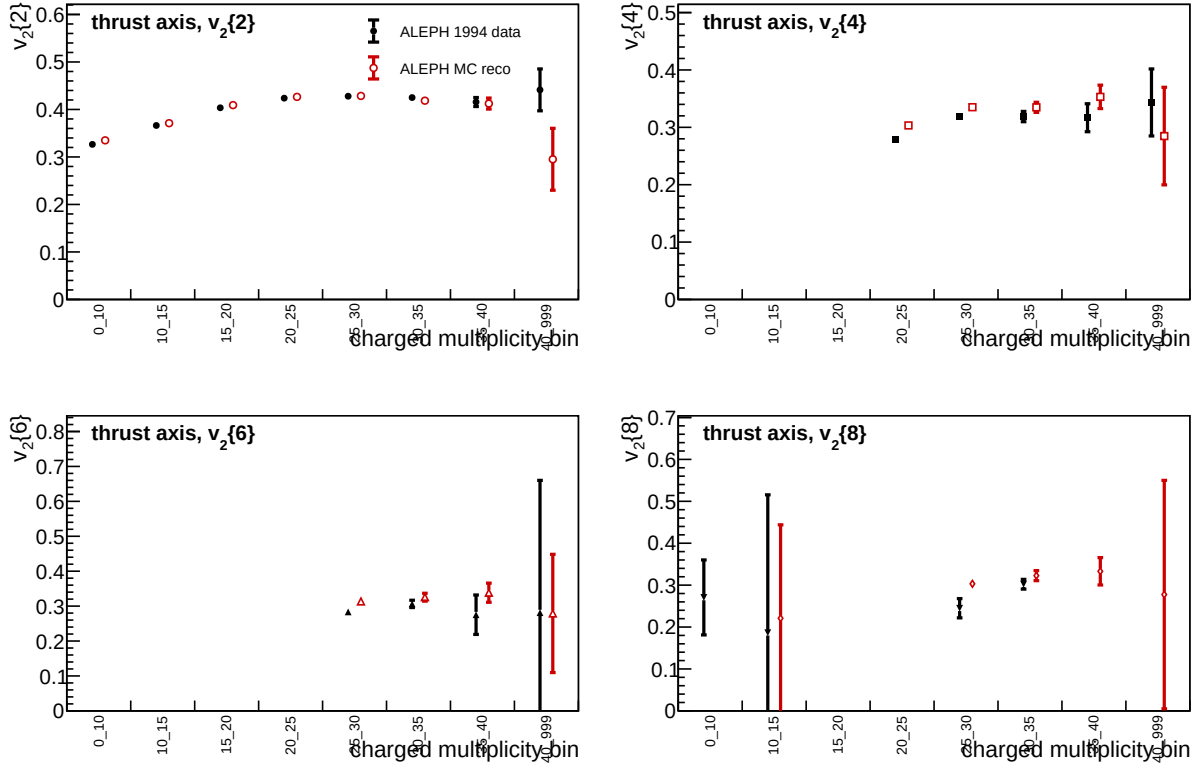


Figure 6: Thrust-axis comparison of matched LEP1 1994 data and reconstructed MC. Missing high-order points correspond to sign-invalid cumulants or bins without a valid real-valued $v_2\{m\}$.

- the current reconstructed-MC comparison is a baseline check, not a detector-response correction or generator systematic uncertainty;
- no variation of the track p_T threshold, charged-particle definition, thrust-axis construction, or event-selection definition is included yet;
- high-order cumulants in the highest-multiplicity bins are statistically fragile and can become sign-invalid.

These items define the next analysis steps before the result can be interpreted as a corrected ALEPH measurement.

8 Summary

A first ALEPH charged-particle cumulant workflow has been built from the local StudyMult code base and validated against exact brute-force ordered-particle sums. The current production processes 59,874 StudyMult events and extracts $v_2\{2\}$, $v_2\{4\}$, $v_2\{6\}$, and $v_2\{8\}$ versus charged-particle multiplicity for both the beam-axis and thrust-axis azimuthal definitions. Statistical uncertainties are propagated through the nonlinear cumulant definitions with a sixteen-chunk jackknife. The beam-axis and thrust-axis figures included in this note provide the current detector-level baseline for subsequent correction and systematic-uncertainty studies.

References

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